

SLIP – 23

Section I: Data Structures [15 Marks]

Q1) Write a menu driven program for the following operations on doubly linked

Q1) Perform the following operations using static implementation of stack i.

Accept n integers from user and push it into the stack. ii. Pop the element from the stack and display it (Use isEmpty())

Answer:

```
#include <stdio.h>
#define MAX 100

int stack[MAX], top;

void initialize()
{
    top = -1;
}

int isEmpty()
{
    return top == -1;
}

int isFull()
{
    return top == MAX - 1;
}

void push(int x)
{
    if(isFull())
        printf("Stack Overflow\n");
    else
        stack[++top] = x;
}

void pop()
{
    if(isEmpty())
        printf("Stack Underflow\n");
    else
        printf("Popped element = %d\n", stack[top--]);
}

void display()
```

```

{
    int i;

    if(isEmpty())
printf("Stack is empty\n");    else
    {
        for(i = top; i >= 0; i--)
printf("%d ", stack[i]);
printf("\n");    }
}

int main()
{
    int n, x, i;

    initialize();

    printf("Enter n: ");
    scanf("%d", &n);

    printf("Enter %d integers:\n", n);
for(i = 0; i < n; i++)
    {
        scanf("%d",
&x);        push(x);    }

    printf("Stack: ");
    display();

    pop();

    printf("Stack after pop: ");
display();

    return 0;
}

```

Section II: Database Management Systems [15 Marks]

Q 2) Consider the following database (Primary keys are underlined)

Customer (cno, cust_name , cust_city)

Account (ano,acc_type , balance)

There exists a One to Many relationships between Customer and Account.

Constraint:Account type Should be either “Saving” or “Current”

Q 2(A)

A.Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```
CREATE TABLE Customer(  
  cno INT PRIMARY KEY,  
  cust_name VARCHAR(50),  
  cust_city VARCHAR(50)  
);  
  
CREATE TABLE Account(  
  ano INT PRIMARY KEY,  
  acc_type VARCHAR(10)  
    CHECK(acc_type IN ('Saving','Current')),  
  balance NUMERIC(12,2),    cno INT REFERENCES  
  Customer(cno)  
);  
  
INSERT INTO Customer VALUES  
(1,'Sahil','Pune'),  
(2,'Martin','Mumbai'), (3,'Kavita','Pune');  
  
INSERT INTO Account VALUES  
(1,'Saving',50000,1),  
(2,'Current',90000,2), (3,'Saving',70000,3),  
(4,'Current',10000,3);
```

Q 2(B)

B.Execute the following queries in PostGreSQL (any 3) [6 marks]

i. Count the number of customers whose name contains substring "in". .

Answer:

```
SELECT * FROM Account ORDER  
BY balance;
```

ii. Increase the balance of all accounts of customer named as "Sahil" by Rs 10000.

Answer:

```
UPDATE Account  
SET balance = balance + 10000  
WHERE cno IN  
(  
  SELECT cno FROM Customer
```

```
WHERE cust_name = 'Sahil'
);
```

iii. Add column "phone_no" to customer table.

Answer:

```
ALTER TABLE Customer
ADD phone_no VARCHAR(15);
```

Q 2(C)

C. Write a query to list the city-wise customer names having minimum balance. [4 marks]

Answer:

```
SELECT C.cust_city, C.cust_name, A.balance
FROM Customer C
JOIN Account A ON C.cno = A.cno
WHERE A.balance =
(
    SELECT MIN(A2.balance)
    FROM Account A2
    JOIN Customer C2 ON A2.cno = C2.cno
    WHERE C2.cust_city = C.cust_city
);
```

Q 3) Viva [5 marks]